AI-Assisted Mathematics Practice and Subsequent Examination Performance: A Reproducibility and Sensitivity Analysis

A reproducible secondary analysis of a classroom mathematics experiment

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Abstract

The purpose of this study was to evaluate the extent to which differences in model specification and statistical inference impact conclusions regarding whether there is a relationship between the use of AI-assisted mathematics practice and improvement in examination performance. This study re-analyzed the publicly available dataset used by Bastani et al. (2025b) which included 2,848 student-session observations of 825 students in 44 classrooms. The original adjusted regression equations reported by Bastani et al. (2025b) were replicated; however, each was subsequently evaluated using five additional types of analysis: (a) equal student weighting, (b) classroom omission, (c) session models, (d) prior-attainment interaction terms, and (e) restricted wild cluster bootstrap inference. There was an association between standard AI access and an improvement of 13.70 percentage points in practice performance; however, standard AI access resulted in an estimated 5.43 point disadvantage on the examination. The estimation of the examination coefficient remained negative across each model specification and when each individual classroom omission was made. However, the p-value based on clustering (.0197) did not agree with the p-value obtained through the use of the bootstrap (.1118); additionally, each model specification that was simplified produced intervals that contained zero. The results indicated that both types of AI provided improved practice performance with no clearly identified advantages on the examination. The findings indicate that there may be a difference between obtaining a similar direction of effect estimates and obtaining statistically significant results. The primary contribution of this study is a reproducibility and sensitivity analysis (i.e., not a discovery of a learning effect, participant replication, or demonstration of mechanisms). Therefore, the findings should be interpreted as being conditional upon the assumptions regarding model selection, sampling, and inferential procedures.

Introduction

Educational researchers have long recognized that AI-assisted successful solutions can provide evidence of supported task completion but do not necessarily provide evidence of the skills available once assistance is removed. Research has established the need to make this distinction. Recently, educational researchers have applied this distinction to generative AI (Yan et al., 2025; Soderstrom & Bjork, 2015) and discussed implications for educational assessment. As such, this study was motivated by existing theory and research; therefore, it provides no new theoretical contributions. Bastani et al. (2025b) conducted a study comparing the effects of standard AI assistance, a safeguarded tutor, and control during classroom-based mathematics practice. They originally reported improved assisted performance and lower subsequent examination performance with the use of standard AI. This study utilizes Bastani et al.'s (2025b) publicly available data and replicates their major coefficients. This study uses the original intervention participants' data and reproduces the headline result(s) in Bastani et al.'s (2025b) work; similarly, this study does not assert that the general ideas presented in Bastani et al.'s (2025b) work concerning examining heterogeneity and robustness differ from those examined by the original

researchers. Instead, this study addresses a narrow question: How robust is the estimated standard AI examination disadvantage across model specifications, classroom influence checks, and inference methods? The previous two items are distinct aspects of estimating an effect. An estimator can obtain a consistent estimate of the direction of an effect (i.e., positive or negative) regardless of whether the resulting estimate reflects a large or small amount of variability. In addition, an estimator can produce an estimate with low variability but still fail to support statistical significance when employing multiple data perturbation methods and/or justifying procedures.

The analysis retains practice outcomes to describe the intervention's immediate benefits, while treating examination inference as its main interpretive focus. It preserves the original four-comparison testing family for transparency rather than redefining that family to favor the revised question. All extensions are retrospective and exploratory. The intended contribution is an auditable account of analytical sensitivity, supported by a verified implementation and bounded psychological interpretation.

Psychological background

Cognitive offloading has been shown to help learners by reducing the amount of mental processing required during problem-solving tasks (Risko & Gilbert, 2016). However, this can be based upon how much of the work is being shifted out of the learner's mind. If the learner is simply given a hint and still needs to come up with his/her own solution, there will likely be less wasted searching time, however, if the learner is provided with a full solution then he/she may not need to engage in the same level of problem solving needed to create their own answer. Therefore, providing a reduced amount of mental effort can result in improved production immediately following training but does not necessarily mean that the learner is able to apply what they learned to solve problems independently at a future date. Because we did not collect data regarding internal mental effort in our study, offloading represents an explanation for the observed differences in the current study but is not empirically supported.

Retrieval practice presents a similar hypothesis. Roediger and Karpicke (2006) showed that actively recalling previously studied information resulted in better long-term retention. Reducing opportunity to recall or generate solutions would likely lead to less retrieval practice and thus potentially impact retention. However, their memory studies did not test the mechanisms of AI systems. Conversely, Cognitive Load Theory provides insight into how certain task requirements can impede learning for novices (Sweller, 1988). More mental effort is not always better – an effective tutor should eliminate irrelevant mental processes while eliminating task requirements that interfere with comprehension.

In addition to possible decreased retention due to lack of retrieval practice, automation bias represents another possible reason for poorer independent performance. Automation bias refers to relying too heavily on automated tools such as AI without properly monitoring their outputs (Parasuraman & Manzey, 2010). There are several factors that contribute to automation bias including: type of task, user familiarity/trust in tool, and system design. While poor post-usage testing scores could indicate a belief in false recommendations, the reanalysis conducted here cannot differentiate between reliance on correct vs. incorrect recommendations or assess whether either factor contributed to the lower score.

Additionally, metacognitive judgments may represent current levels of fluency (Bjork et al., 2013), rather than demonstrating mastery of material. Users may report smooth interactions as productive/educational regardless of their ability to perform independently. Buçinca et al. (2021) demonstrated through cognitive forcing interventions that users' inappropriate reliance on AI can decrease, with subjective trade-offs. Lee et al. (2025) demonstrated associations between users' confidence in AI and self-reported critical thinking. Survey findings do not equate with cognitive decline.

Each account motivates selecting appropriate metrics to evaluate outcomes. Additionally, none of these accounts support treating all metrics of productivity, perceived effort, recommendation acceptance, and exam scores as equivalent representations of independent thought.

Methodology

Study data origin and scope

The PNAS publication corrected an author affiliation error not impacting any aspect of the research (PNAS, 2025). The original data were publicly accessible via the authors' GenAICanHarmLearning repository accessed October 17th, 2026 at commit 2f63dae1a01d51453826fe07ef5cf6678e339588. Analysis utilized main_regressions/final_data.csv and examined the main_analysis.R script documentation along with supporting documentation (Bastani et al., 2025a). The original study represented a series of math lessons presented to students in a Turkish High School. Three conditions existed in the original study: a GPT-based AI assistant, a tutor with built-in learning safeguards, and control. Students completed the lesson followed by an independent examination without access to any technology. Class assignments and procedural details were originally documented by Bastani et al. (2025b).

The present manuscript utilized pre-existing anonymous records. As such, there was no participant recruitment or contact and no new intervention occurred. Rather than assessing clinically valid outcomes, the present manuscript examines an educational outcome. Student IDs were used solely for counting the number of unique students and examining the structural integrity of each record. Student IDs were never visible within the present manuscript. Instead, reproducibility materials include links to the source dataset as well as its checksum rather than distributing participant records.

Sample construction

Upon downloading the file containing 3,255 observations representing 943 unique students enrolled in 50 classes; no duplicate student-session pairs were identified. Following the methodology employed in the source manuscript, the first phase of sampling excluded honors classes by utilizing Honors = 0. This produced a sample size of 2,899 observations from 839 students enrolled in 44 classes. The decision to exclude honors classes was made because honor classifications differ from those assigned for regular classes in the original design.

The subsequent modeling phases employed a single complete-case sample for both dependent variables. Variables necessary for model estimation included: the two performance assessments, treatment group indicators (i.e., Control/Tutor/AI), prior academic achievement (prior GPA), teacher, session, grader, school year, and class. Fifty (50) non-honor observations contained missing prior GPA values. Only one additional non-honor observation had a missing grader value. These exclusions did not overlap, leaving 2,848 observations from 825 students in 44 classes. Neither outcome was missing in the retained non-honor file. The above counts represent available datasets as opposed to documenting complete attendance throughout each session.

Ultimately, the dataset consisted of 1,062 control observations, 824 standard AI-assisted observations, and 962 safeguarded tutor-assisted observations. Similarly, unique-student counts for each condition were 316 for Control, 238 for Standard AI-Assist, and 271 for Safeguarded Tutor-Assist. Finally, class counts were 17 for Control, 13 for Standard AI-Assist, and 14 for Safeguarded Tutor-Assist. Employing a single sample facilitates interpretation across both dependent variables. Complete

case analysis remains contingent upon the missing data process; absent a score does not imply that exclusion occurred due to missingness alone

Table 1

Complete case sample and observed performance

Condition	Students	Classes	Records	Practice M (SD)	Exam M (SD)
Control	316	17	1062	28.47 (28.81)	32.38 (27.72)
Standard AI	238	13	824	47.49 (25.65)	28.93 (23.30)
Safeguarded tutor	271	14	962	67.00 (30.92)	31.87 (28.17)

Note. Scores are percentages. Records are student-session observations. Means and standard deviations are unadjusted; repeated observations are not independent participants.

Variables and estimands

A 0-to-1 scaled version of Part2Tot measured practice performance, and a 0-to-1 scaled version of Part3Tot measured examination performance. When multiplying a coefficient by 100, this calculates an absolute difference in terms of percentage points. Both GPTBase and GPTTutor represented treatment indicators, with both = 0 for control. Previous GPA was entered as a covariate variable. All teacher, session, grader, and school year variables were treated as categorical variables, matching the source script. Including them in the model allows for comparisons to adjust for any pre-existing differences associated with these categories.

The main estimands were the two treatment-versus-control coefficients for each of the two outcomes. Since these represent assignment-based comparisons, they do not restrict the estimates to students that choose to utilize the assistant intensively. In fact, actual utilization may depend on either the students' motivation or difficulty utilizing the tool. Therefore, comparing heavy users to light users could be difficult to interpret causally. As stated above, since this re-analysis replicates the original specifications, the identification assumptions also replicate those of the source study and its assignment mechanism. Thus, simply reproducing the coefficients will not provide verification of the assignment mechanism.

Statistical analysis

Using separate ordinary least-squares regression equations predicted practice and examination scores from the treatment indicators, GPA, and category controls. Clustered standard error estimates were provided at the class-level allowing for correlated errors within each class including correlations across multiple session observations. This is generally better than treating each observation as independent given that support is typically allocated by class (Cameron & Miller, 2015). The finite-sample multiplier for the variance-covariance estimation is G / (G – 1) * ((N – 1) / (N – K)) where G represents the number of classrooms, N represents total number of student-session observations, and K represents the rank of the fitted design matrix. Two-sided tests and 95 % confidence intervals were generated using a t-distribution with G – 1 degrees of freedom providing 43 degrees of freedom in the primary models. There were four primary comparisons made simultaneously: the treatment group (both standard AI and tutor) compared to control for each of the two outcomes. Holm adjustment was applied to their respective p-values (Holm, 1979). Tutor versus standard AI comparisons were exploratory comparisons and are presented without adjusting for family-wise error. This is defined retrospectively as well as explicitly; it does not turn a secondary analysis into a preregistered confirmatory test.

There were two sensitivity analyses conducted to fit both outcomes. The first sensitivity analysis added honors classes to the sample and retained all complete records from the original data set. The second sensitivity analysis removed class-session combinations that were determined to have been non-

compliant based on the original analysis script. None of these sensitivity analyses constitute new experiments. The first alters the population studied, whereas the second uses information regarding compliance as a condition for which records are included. Together they determine if the numeric patterns are largely dependent upon such factors or not.

The implemented Python code utilized libraries NumPy, pandas, SciPy, Statsmodels and Matplotlib. A fresh source download verified its integrity via a checksum; eight numerically equivalent regenerated CSV files confirmed prior output differences within a tolerance of 1E-10. Additionally there were twenty-two checks covering sample characteristics, matrix rank, primary coefficients and clustered uncertainty measures; Holm adjustment; bootstrap statistics; and decisions related to equivalence. Cross-checks of primary calculations were performed with Statsmodels, and forty-eight bootstrap statistics were individually recalculated. The above mentioned checks validated that the implementation of numbers was correct and not causal validity. The optional R companion was not run due to R being unavailable.

Expanded analytical checks

All expanded checks were selected post-publication knowledge of the final results and thus are exploratory in nature. There are four specifications that use identical complete case datasets: Treatment Indicators Alone; Treatment Plus Prior GPA and Session Indicators; Full Source Specification; and Full Specification Weighted By Reciprocal Of Observed Number Of Sessions Per Student. The weighted model assigns each observed student an equal amount of weight regardless of how many sessions he/she has attended. This changes how the weighting of the estimand is constructed and does not recover outcomes for missing students. Holding the dataset fixed allows differences among specifications to reflect adjustments to covariates rather than changes in included records.

Classroom Influence was examined by iteratively dropping out one class at a time while refitting both full models. The resulting 44 estimates per treatment/outcome demonstrate whether or not removal of one class reverses direction of the estimated effect. Note that overlapping fits such as these are not independent replications nor is the range indicative of a confidence interval. Generalized-inverse least squares is employed to handle redundant nuisance indicators when fitting after subsetting.

Inference regarding examination results was further evaluated using a Restricted Wild Bootstrap procedure with Rademacher Weights; 4999 Draws; Random Seed 20261017. Separate bootstraps were performed for each treatment coefficient. Each bootstrap consisted of generating null model fitted values + residuals multiplied by one random sign per classroom to generate bootstrap outcomes. Bootstrapped draws were then refit and studentized using the classroom-clustered covariance estimate. The two sided p-value counts number of bootstrap t-statistics with magnitude greater than or equal to that of the observed statistic [with a plus one correction]. The above described method is a model-based check and NOT a form of randomization inference. Model-based bootstrap procedures address issues related to cluster approximations but can themselves be sensitive to aspects of cluster structure (MacKinnon & Webb, 2017).

Moderated regression models allowed for interactions between both the treatment indicators and grand-mean centered previous GPA while retaining all of the covariates from the full model. Narrative coefficients have been scaled to reflect a 0.10 unit increase in recorded GPA. Each session has separate specifications of the full model. However, these comparisons do not establish causality between prior attainment or cumulative AI exposure. There is no multiplicity correction applied to the exploratory comparisons listed here. Therefore, they are pointwise and p-values may not represent confirmatory evidence.

Using full model estimates, clustered standard error, and 43 degrees of freedom, two one-sided tests examined whether there were differences in examination equivalence at two, three, and five percentage points. For alpha = .05, if the 90% CI lies completely within the selected range, then both one-sided tests reject at alpha = .05, supporting equivalence within the defined margin. Because the selection of these ranges was retrospective and without regard to what constitutes an educationally significant margin of equivalence, this represents a sensitivity analysis based upon Lakens' (2017) distinction between a non-significant difference and equivalence.

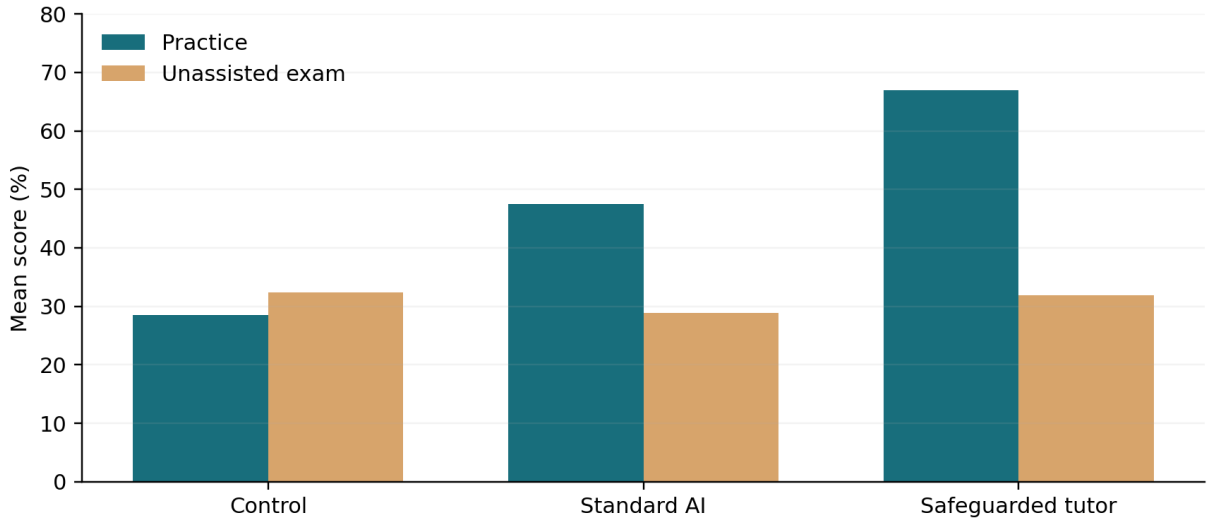
Results

Observed scores and adjusted comparisons

Table 1 provides descriptive means for the same complete-case sample used for the original regressions. The mean score for practice was 28.47% in control, 47.49% with standard AI and 67.00% with the AI Safeguarded Tutor. The average examination scores were 32.38%, 28.93%, and 31.87%, respectively. These represent raw averages and have not been adjusted by covariates. While providing a useful way to describe the record data, these numbers should never replace the adjusted estimates.

Figure 1

Observed scores in the complete case sample



Note. Unadjusted means; practice and exam tasks differ. Bar heights do not represent within-person learning loss.

The standard AI practice coefficient was .1370 or 13.70 percentage points greater than control. The 95% CI for this estimate spanned from 7.45 to 19.95 points. The practice coefficient for the tutor was .3613 or 36.13 points better than control. The corresponding 95% CI for this estimate spanned from 29.73 to 42.53 points. Both of these estimates were significantly different from zero ($p < .001$), even when adjusting for all four primary comparisons.

As a whole, the practice model accounted for about 38.9% of the variation in the observed practice scores among participants given treatment and accounting for all covariates.

In contrast, examination results did not follow the same pattern. The coefficient for the standard AI was -0.0543 or -5.43 percentage points less than control. The 95% CI for this estimate spanned from -9.96 to -0.91 points. Before adjusting for multiple testing, this result was statistically significant ($p = .020$). After adjusting for multiple testing via Holm’s method, however, this result remained statistically

significant at .05 (p = .039). As a result, although there was evidence that the standard AI resulted in reduced examination scores relative to control (at least in terms of statistical significance), the exact size of this reduction cannot be determined from the estimated interval.

Likewise, the coefficient representing examination performance under the AI Safeguarded Tutor was -0.0043, which translates into a -0.43-point disadvantage compared to control and has a 95% CI spanning from -3.13 to 2.27 points. Although the estimated disadvantage associated with the AI Safeguarded Tutor relative to control is small (-0.43 points) and of uncertain practical importance, it is difficult to assert that there was a benefit of this program over control. Likewise, there is no clear indication of whether this program was equivalent to control. Based on the information available, we cannot rule out modestly sized positive or negative effects associated with this program. Finally, the R-squared statistic associated with the examination model indicates that about 38.6% of the variance in observed examination scores could be predicted from a combination of treatment indicator(s) and covariates.

Table 2

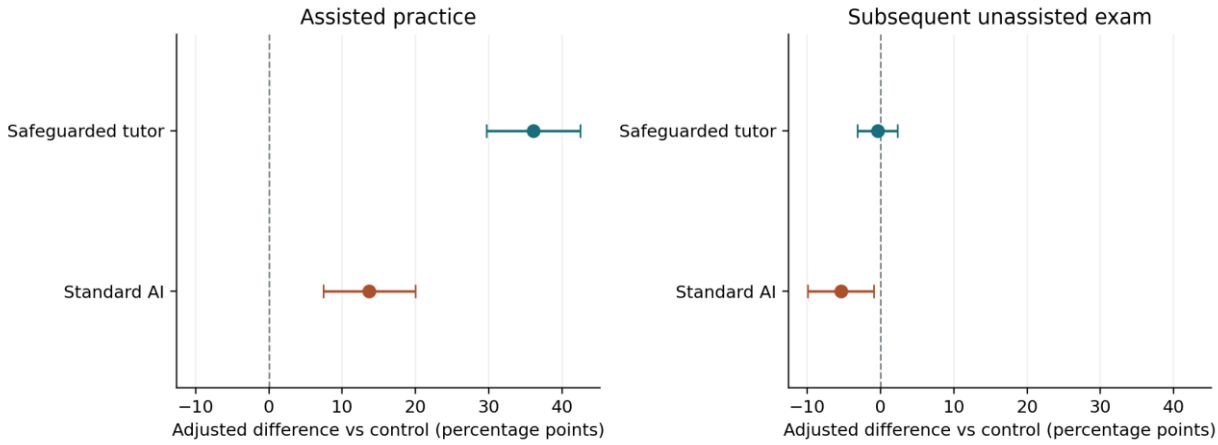
Primary adjusted comparisons with control

Outcome	Condition	Difference	95% CI	p	Holm p
Practice	Standard AI	13.70	[7.45, 19.95]	< .001	< .001
Practice	Tutor	36.13	[29.73, 42.53]	< .001	< .001
Exam	Standard AI	-5.43	[-9.96, -0.91]	0.020	0.039
Exam	Tutor	-0.43	[-3.13, 2.27]	0.749	0.749

Note. Differences and intervals are in percentage points versus control. Each model uses 2,848 records and 44 classes. Confidence intervals are pointwise. Tests use 43 degrees of freedom. Holm adjustment covers the four rows shown.

Figure 2

Adjusted effects on practice and unassisted examination performance



Note. Dots show adjusted differences versus control. Bars show pointwise 95% confidence intervals using classroom-clustered standard errors and 43 degrees of freedom. The dashed line indicates no difference.

Exploratory direct comparison

The estimated adjusted difference in examination performance between the AI Safeguarded Tutor and the standard AI programs is 5.00 percentage points (95% CI = [0.30, 9.70], p = .038). In contrast, the estimated difference in practice scores is 22.43 points (95% CI = [15.53, 29.32]). These estimates suggest that the two programs resulted in different levels of improvement in practice versus examination scores, but we cannot identify how much of this improvement was due to specific components of each program since each condition represented a package of interventions. Because examining whether there is a

difference in examination performance between these two programs is an exploratory analysis, caution should be exercised regarding any conclusions drawn from this estimate. Although neither of the standard AI examination comparison nor the direct tutor vs standard AI comparison would be significant if all six primary model comparisons were considered as part of one Holm-adjusted family (i.e., adjusted p-values would approximate .059 and .075), these results demonstrate how sensitive these estimates are to being part of a larger testing family. To avoid placing too much reliance on arbitrary decision- making rules (i.e., "is it significant?"), it might be beneficial to report effect size estimates along with CIs as opposed to relying on p-value labels.

Sensitivity analyses and numerical reproduction

The inclusion of students enrolled in honors classes provided 3200 complete cases across 50 classes. The standard AI coefficients were 10.15 points for practice and -6.39 for examination; tutor coefficients were 35.90 and 0.14 points, respectively.

Removing the reported non-compliance cases removed some class-session observations and reduced observation counts to 2805 across 44 classes. The standard AI coefficients for practice and exam were now 13.69 and -5.41 points; tutor coefficients were 36.19 and -0.44 points, respectively. These analyses maintained the general contrast between improved practice performance and lack of an examination benefit relative to control

Table 3

Sensitivity of adjusted treatment coefficients

Sample	Records	Classes	Practice Base	Practice Tutor	Exam Base	Exam Tutor
Primary	2848	44	13.70	36.13	-5.43	-0.43
Including honors	3200	50	10.15	35.90	-6.39	0.14
Excluding noncompliance	2805	44	13.69	36.19	-5.41	-0.44

Note. Coefficients are percentage-point differences versus control. Base denotes standard AI. These sensitivity analyses are descriptive robustness checks; full intervals are supplied in regression_results.csv.

As indicated by the primary treatment coefficients being equal to the published values of .137, .361, -.054, and -.004 to 3 decimal place, there was support for the correct implementation of the data handling and fitting of the specification. However, such agreement did not provide additional (independent) evidence based on data collected from other schools or samples. The confidence intervals and explicitly stated testing adjustments added detail to a previously reported analysis; they should not be considered as a second study when developing a synthesis of the literature.

Model specification and student weighting

Table 4

Examination contrasts across model specifications

Specification	Condition	Points	95% CI	p
Unadjusted	Standard AI	-3.45	[-10.22, 3.33]	0.311
Unadjusted	Tutor	-0.51	[-8.47, 7.46]	0.898
Session and GPA	Standard AI	-2.48	[-8.58, 3.62]	0.417
Session and GPA	Tutor	0.65	[-6.01, 7.31]	0.845
Adjusted	Standard AI	-5.43	[-9.96, -0.91]	0.020
Adjusted	Tutor	-0.43	[-3.13, 2.27]	0.749
Equal student weight	Standard AI	-5.29	[-9.69, -0.89]	0.020
Equal student weight	Tutor	-0.49	[-3.35, 2.37]	0.731

Note. Same 2,848 records throughout. Full adjustment includes previous GPA, teacher, session, grader, and year. P values use conventional classroom clustering. All extensions are exploratory.

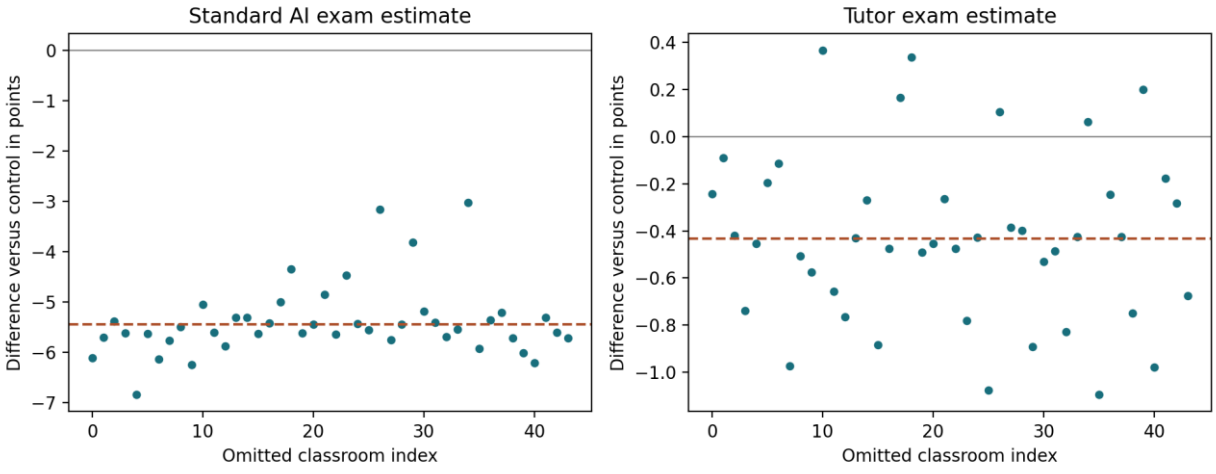
All four specifications produced positive practice advantage. The unadjusted standard AI comparison of examinations had a -3.45 point disadvantage, with a 95% confidence interval ranging from -10.22 to 3.33 and p = .311. Adjusting for both GPA and session resulted in a -2.48 point disadvantage, with a 95% CI ranging from -8.58 to 3.62 and p = .417. The fully adjusted model provided a -5.43 point disadvantage and p = .020. As the point estimate was consistently negative across each of the specifications however, conventional statistical significance was dependent upon the full covariate adjustment. Thus, stating the evidence as statistically robust to all reasonable model selections would represent an exaggeration of these findings.

Standard AI exam estimates using equal total weight per observed student produced an estimate of -5.29 points with a 95% CI ranging from -9.69 to -0.89 and p = .020. Tutor exam estimates produced an estimate of -0.49 points with a 95% CI ranging from -3.35 to 2.37. The similar estimates do not suggest that differential numbers of observed sessions drove the full model results.

Classroom influence and bootstrap inference

Figure 3

Examination estimates after removing each classroom



Note. Dashed lines show full-sample coefficients. Each point omits one classroom; the range is not a confidence interval.

Across the omitted classrooms (total of 44), standard AI exam coefficients varied from -6.83 to -3.02 points. Their signs never reversed. Tutor exam coefficients varied from -1.09 to 0.37 points. Practice coefficients were always positive: 11.90 to 16.35 points for standard AI and 34.13 to 39.18 for tutors. The full model pattern did not rely solely on one classroom; however, the magnitude of each class affected the size of the estimated coefficient.

Restricted wild cluster bootstrap p-values were .1118 for standard AI and .7916 for the tutor on the exam. The Monte Carlo standard error of the first p-value is about .0045; however, this is less than what explains its difference from the traditional clustered p-value of .020. Although both methods utilize the negative coefficient, they have different reference distributions for estimating uncertainty. Thus, the methods weaken the case for a binary statement that the effect has been statistically established as harmful; however, they do not establish that the effect is non-existent.

Bootstrapped inference addresses a different vulnerability (i.e., approximating the distribution of the test statistic); thus, classroom omission examines potential influence while bootstrap inference examines an alternative approximation to the null reference distribution. Negative estimates may remain stable to individual classroom removals; however, the statistical significance of these estimates remains subject to the choice of inference method. Neither check adds new participants nor confirms through independent experiments.

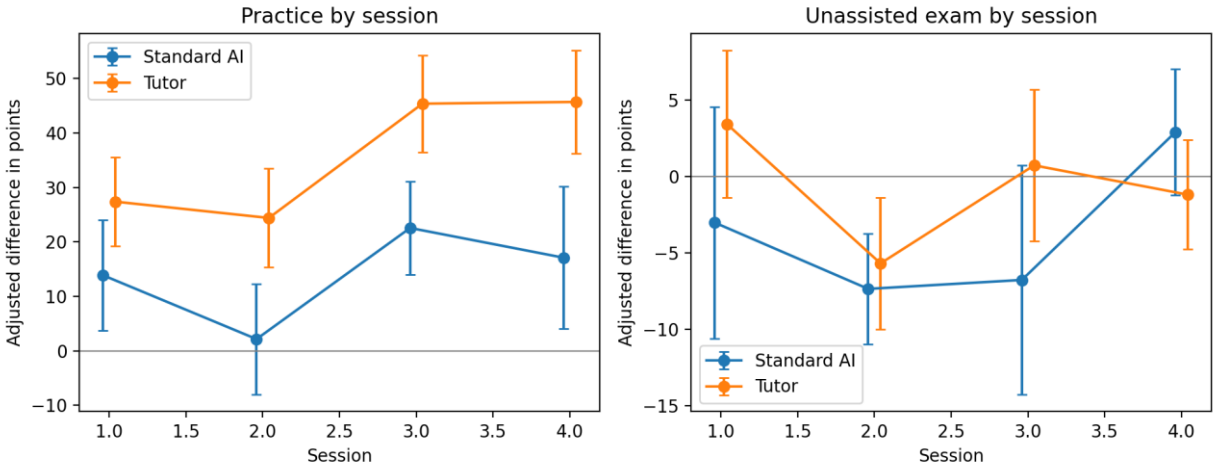
Prior attainment and instructional sessions

An increase in prior GPA of .10 units was related to smaller standard AI practice advantages of 6.08 points; with a scaled 95% CI ranging from -9.67 to -2.50 and p = .001. Correspondingly, prior GPA increases were also related to smaller tutor practice advantages of 4.18 points; with a 95% CI ranging from -7.74 to -0.62 and p = .022. Exploratory slope estimates suggest that larger practice advantages exist among students who have lower prior attainment levels; ceiling effects, task difficulty, and differences in helpfulness of assistance are plausible explanations that these models cannot distinguish.

Exam interaction estimates were less conclusive. Interaction per .10 GPA unit for standard AI exams was -2.33 points; with a 95% CI ranging from -5.11 to 0.45 and p = .098. Interaction per .10 GPA unit for tutor exams was -0.40 points; with a 95% CI ranging from -3.50 to 2.70 and p = .795. Therefore, even though larger practice advantages among students who have lower GPA exist; these estimates do not demonstrate that students who have lower GPA levels learn more independently than students with higher GPA levels. Subgroup estimates require further prospective testing before consideration of student specific application

Figure 4

Adjusted treatment contrasts across sessions



Note. Bars show pointwise 95% intervals with classroom clustering. Session models are exploratory and do not identify a causal exposure trajectory.

It may have occurred to you to wonder what differentiates the examinations taken by students during their participation in the AI Supported Mathematics Intervention from each other. Each of the first four sessions produced a standard AI examination difference compared to control in the following order: -3.01, -7.36, -6.77, +2.91 percentage points. Each of the first four sessions produced a tutor examination difference compared to control in the following order: +3.43, -5.69, +0.73, and -1.18 percentage points.

The only time when a negative examination estimate was found for each of the two interventions was at session two. The only time that the standard AI supported mathematics estimate was not negative was at session four.

The pooled disadvantage does not reflect all occasions. Differences in the content, assessments, attendance, and composition of estimable contrasts preclude any assumption that this represents a simple trend of learning loss or recovery. The results of the estimates for each session along with the corresponding point-wise intervals are contained within the package.

Equivalence and acceptable margin

The 90% interval around the tutor's examination estimate was (-2.68, 1.82). This equated to no equivalence being established at a margin of +/- 2 percentage points (TOST $p = .124$) but did equate to establishing equivalence at margins of +/- 3 ($p = .031$) and +/- 5 ($p < .001$) using the conventional clustered inference technique. Standard AI failed to meet all three margins. Therefore, the tutor can only be considered equivalent with regard to a sufficiently broad acceptable margin and specifically in relation to this particular type of inference technique. These post-hoc tests do not demonstrate that the intervention is educationally harmless based upon every conceivable definition.

Discussion

Contribution of analytical sensitivity

The replicated standard AI-supported examination coefficient indicates a negative association between standard AI access and subsequent examination performance. However, the degree of evidence supporting the comparison is contingent upon how the comparison is modeled and tested. Specifically, a

fully adjusted model produces a p value less than .05 according to conventional clustered inference techniques. Less complex models and use of a restricted wild cluster bootstrap fail to produce such a result. Omitting individual classrooms preserves the negative sign; giving students equal total weight closely replicates the full model estimate. Thus, there appears to be some distinction in terms of the findings presented here regarding stability of estimated direction vs. stability of inferential confidence. Notably, failure to report both types of robustness check would hide this distinction.

The bootstrap does not alter the observed coefficient. Rather, it alters the reference distribution used to evaluate whether the studentized coefficient falls outside of some portion of the null distribution. Therefore, any disparity between the conventional methods employed in this investigation and the bootstrap methods employed herein represent uncertainty issues, and not alternative measures of direction of effect. Additionally, despite providing data from forty-four classrooms (i.e., fewer independent allocation units than suggested by the 2,848 rows of data), cluster size, leverage, and the relationships among nuisance covariates and treatment could account for discrepancies in approximation quality. While these are plausible explanations for why such investigations are warranted, they represent possibilities for investigation and not established diagnoses. Neither method should be selected because it yields a preferred conclusion. Both conventional clustered inference techniques and the selected bootstrap methods assume certain things about data generation processes; however, this manuscript does not establish which method possesses better finite sample calibration properties with respect to the original allocation structure. Further studies investigating additional comparisons to established bootstrap methodologies; procedures incorporating adjustments for leverage; and randomization inference methodologies that honor experimental design features would increase a methodology-focused assertion. At minimum, sufficient justification exists to include disagreements rather than assert either that harm has been demonstrated or dis-proven.

Substantive interpretation of specification sensitivity

The specification sensitivity of the comparison between treatments and control group reflects a substantive difference between the comparisons made by unadjusted analyses versus those made through multiple regression modeling. Unadjusted comparisons describe differences between observable treatment groups prior to adjusting for recorded instructional circumstances. Multiple regression modeling adjusts outcomes for prior grade-point average, teacher effects, session number, grader effects, and school-year effects in addition to comparing treatment groups. Stronger negative standard-AI-versus-control associations exist in the fully adjusted regression model; however, these stronger associations do not necessarily indicate bias nor do weaker standard-AI-versus-control associations in unadjusted analyses necessary imply that unadjusted analyses are preferable. Covariates can help to improve precision associated with estimating treatment effects or address imbalance in measured attributes; however, covariates can also influence how information contributes to estimating treatment effects. Justification for including covariates should be derived from both the design and measurement process rather than solely from significance labels generated by statistical testing.

The same rationale applies to the increased focus on examining inference. Revising emphasis clarifies contributions rather than creates prospective hypotheses or diminishes awareness of existing literature. Maintaining visibility of original four-test families allows readers to assess estimated differences, conventional intervals surrounding estimated differences, bootstrap-tail probabilities surrounding estimated differences, and sensitivity models collectively.

A more compelling future analysis would specify a priori how disagreement among procedures will be treated and validated those procedures through simulations reflecting actual allocation structures. That

would allow researchers to treat calibration as a direct test rather than viewing agreement with a preferred outcome as validation.

Position relative to earlier research

The distinction between AI-supported performance and learning is already established. Yan et al. (2025) discuss it directly in a psychological commentary. Kovari (2026) provides a related framework and retrospective analysis emphasizing assessment conditions. The present study does not introduce that conceptual distinction or a new statistical method. Its narrower contribution is the explicit comparison of specifications and inference procedures in a reproduced mathematics analysis. No priority claim is made for this exact combination of checks.

Table 5

Related research and its relationship to this analysis

Study	Design or context	Measured outcomes	Interpretive limit
Bastani et al. 2025b	Classroom mathematics	Practice and unassisted exam	Source of this reanalysis
Fan et al. 2025	117 university students	Essay quality, knowledge, transfer	Better essay scores do not establish transfer
Gajos and Mamykina 2022	Three decision support experiments	Decision accuracy and incidental learning	Explanation only differs from answer plus explanation
Melumad and Yun 2025	Seven search experiments	Depth and originality of subsequent advice	Advice measures differ from mathematics tests
Kestin et al. 2025	University physics tutoring	Assessed learning versus active class	Designed tutor and different comparator
Noy and Zhang 2023	Professional writing experiment	Assisted quality and speed	Independent retention not established
Yan et al. 2025	Psychological commentary	Performance versus learning	Conceptual distinction predates this analysis
Kovari 2026	Framework and secondary analysis	Assessment conditions	Related framing predates this analysis

Note. Experimental and conceptual sources are distinguished. Additional studies are compared qualitatively; no pooled effect or participant data reanalysis is claimed. The scope of Kovari was checked against the author repository.

Fan et al. (2025) randomly assigned 117 university students to write with different forms of support, including a chatbot. While their results indicated higher grades when writing with a chatbot than when writing alone, there were no statistically significant differences in what the students knew and could apply. In addition to measuring the products of writing processes, the authors' study helped to determine whether students learned through writing with the chatbot versus if they simply relied upon the chatbot for answers. However, even though there were no statistically significant differences found regarding the ability of students to apply knowledge gained, this does not necessarily mean that students who wrote with the chatbot did learn as much as students who wrote on their own.

Similarly, Gajos and Mamykina (2022) distinguished between immediate decision accuracy and incidental learning in their study. The authors explained that participants who made decisions based solely on information provided in an explanation-only condition demonstrated an increase in learning. Therefore, how people use information in order to answer questions appears to be equally important to whether or not people receive an explanation.

Melumad and Yun (2025) conducted a series of seven studies comparing LLM-generated summaries to web searches. Melumad and Yun stated that accessing information through LLM-generated summaries led to shallower subsequent recommendations as well as providing less detail and fewer new ideas. As

opposed to students taking a mathematics exam, however, Kestin et al. (2025) reported greater benefit for students utilizing a deliberately created university physics tutoring system than students utilizing an active-learning approach. Similarly, Noy and Zhang (2023) found increased productivity benefits from assisted writing without demonstrating long-term independent retention. Each of these studies makes universally applicable claims about AI and learning highly unlikely; each has failed to provide directly interchangeable estimates of effects.

On October 17, 2026, I completed a focused critical update of the literature on 17 October 2026 that combined generative AI, independent performance, cognitive engagement, and learning, and then verified the author's and publisher's record. This is not a comprehensive systematic review of all studies related to generative AI; this is a focused critical comparison of selected studies. Although the publisher allowed me to preview Yan et al., due to my inability to access the entire Kovari article via the publisher I had to verify the placement of Kovari based on his author-created repository. I am claiming neither participant-level re-analysis nor pooled effect of the additional studies included.

Educational implications and psychological interpretation

An informative hypothesis suggests distinguishing help that eliminates obstacles to learning from help that ultimately completes the operation being assessed. Hinting can preserve computation; providing complete solutions can replace it. Providing source summaries can enhance accessibility at the expense of eliminating the need for comparison and integration of sources. These theories suggest measurable processes, e.g., independent attempts, revisions, retrieval, and verification. These theories are consistent with the findings presented herein but cannot be inferred as causal mechanisms based solely on examination coefficient values. Practically speaking, students should be evaluated after assistance terminates while keeping supported performance separate. These analyses do not establish that intelligence will be lost or that critical thinking will decline over time. Lower scores immediately post-assistance may indicate weaker acquisition, differences in engagement, interference or other processes. Larger advantage in terms of practice among lower-GPA students does not demonstrate larger independent learning gains. Session variability further complicates drawing conclusions about a universal disadvantage across instructional sessions. Equivalence of tutors depends on an acceptable margin as opposed to merely a lack of statistical significance.

A discriminating next experiment

Randomizing classrooms into three conditions would allow for controlling the type of instructional tool utilized (control, direct-answer AI or hint first AI), ensuring model consistency, verified content consistency, and maintaining equivalent time budgets throughout. The hint-first condition would require students to make an initial attempt prior to revealing a solution. A delayed unassisted transfer assessment one week later would serve as the primary outcome; secondary outcomes would include immediate independent performance, supported accuracy and process measures. This would isolate a particular instructional technique from differences associated with distinct whole tutoring packages.

Prior to having access to outcome data, registration of exclusions, testing family, equivalence bounds, and inference procedures must be documented. Planning sample sizes must take into consideration classroom level variability, intraclass correlation, potential for attrition, and a minimally educationally relevant difference. Blind scoring and parallel forms would add strength to assessment methods. Additionally, a component experiment could examine manipulating number of attempts and/or availability of answers.

These represent proposed experimental modifications that were not implemented for this manuscript. This manuscript represents a secondary analysis of previously collected data, thus retaining reproducibility and transparent uncertainty as its most substantial contributions.

Limitations and future research

Each numerical estimate originates from only one previous experiment. Secondary analyses can illuminate assumptions and reporting issues, yet cannot expand the size of the participant population, settings, or instructional materials. Studies examining other academic subjects, adult workplace environments or later generations of AI will necessitate collecting additional data. The original experiment(s) and designs remained those developed by Bastani and colleagues; this manuscript represents computational reproduction and expansion of that experiment -- NOT an independent participant replication.

The dispute concerning which inference method is preferable remains unresolved. Omitting classroom records assesses influence but do not confirm both null distributions. The bootstrap utilizes one set of weights and one manner of constructing residuals -- AND NO reconstruction of the original assignment mechanism was undertaken. Numerical agreement with separate library fittings validates that implementation occurred correctly -- NOT that one method provides superior calibration to the other method. Future methodological benchmarking efforts should occur prior to making a definitive assertion that one test is superiorly calibrated relative to the other test

Complete-case selection excluded 51 non-honors records, and equal student weighting cannot recover missing attendance or remove selection bias. Including honors changes the population rather than repairing missingness. The analysis also does not measure delayed retention, general critical thinking, trust, cognitive load, or a mediated psychological pathway. Examination scores should not substitute for these constructs.

Finally, the decisions made analytically were all post hoc (i.e., after the fact), and the exploratory session, interaction, and equivalence analyses did not have one single pre-specified multiplicity adjustment. Additionally, the literature review conducted in this study was "targeted" as opposed to being a "systematic" review. These methodological considerations support the potential for a focused reproducibility contribution versus making general theoretical contributions. Furthermore, a second compatible dataset with a prospectively specified validation would be better evidence than conducting additional unplanned tests using the same data set.

Conclusion

This re-analysis demonstrates that it is possible to reproduce both the positive and negative unstandardized effects of AI-based student performance measures and the original AI-based student performance measure coefficients found by Bastani et al. (2025b). However, these findings also indicate that the amount of confidence in inference varies based upon the analytical options selected. Specifically, the results demonstrate that regardless of which compared specification is used or individual classroom is omitted; the negative standard AI examination estimates remain consistent. However, when comparing the conclusions drawn using the conventional clustered statistical approach vs. restricted bootstrap inference at the .05 alpha level; they yield different conclusions. Additionally, the results clearly show that there is no detectable examination benefit to the use of a safe-guarded tutoring strategy, and that whether or not equivalence can be demonstrated depends on how large of a margin is selected and the inference method. Therefore, although this study contributes to the educational literature by providing a

transparency analysis regarding the sensitivity of short-term exam results to various analytical options; it does so without providing any evidence of a new general learning effect. As such, educational evaluations need to differentiate between students' performance on their own work as compared to their performance on tasks where they receive assistance through some form of technology or other external resource; and educational evaluators need to take into account and reflect in their substantive findings the degree of uncertainty associated with each justified analytical option.

Data and analysis availability

The original dataset and source analysis belong to Bastani and colleagues and are publicly available at https://github.com/obastani/GenAICanHarmLearning. The accompanying reproducibility package includes the executed Python analysis, a companion R script, generated summary data, figures, and instructions for retrieving the pinned source file. The primary coefficients were checked against the published table. This manuscript is an unpublished independent research report and a secondary analysis of existing data.

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Appendix A Reproducibility details

The source file for the data was a .csv file (final_data.csv) located in the "main_regressions" directory within the authors' repository. The SHA-256 hash value of the data is: 430055b4633eaaf87879bb38eb4b1c9425b91e1d7f529029dc1883e33d91df0a. This ensures that when performing the analysis, it will utilize the exact same byte sequence as when generating the hash code, to prevent using a version of the data which has been modified without documentation.

The design matrix used in each regression included 35 linearly independent columns (including the intercept), 2,848 observations, and 44 classrooms. Both of the dependent variables utilized identical control variables. Class-level sum of product of regressor and residual values, plus a finite sample correction described in the Methods Section were used in constructing the covariance estimator. Confidence intervals around point estimates were not corrected for simultaneous coverage.

In addition, there were several (non-compliant) combinations of class and session numbers which were specifically excluded from being analyzed with respect to their sensitivity model: class 11S in Sessions 2 & 4, class 10A in Session 2, class 9J in Session 1, and class 10B in Session 3. Additionally, because of how honors students were excluded from the original study samples, there have been some combinations removed from the original complete-case sample. Thus, the total number of student-session observations represented in the sensitivity model was reduced by 43 compared to the primary complete-case sample.

It is important to note that prior to removing missing covariates, the total number of students in the non-honor group was 839; however after removing missing covariates and reducing the sample size to include only those who had complete cases on all variables for the analyses, the total number of students in the non-honor group was reduced to 825. Similarly, the four treatment effect coefficients reported in the publication represent what can be described as "reproduced" results; whereas, the exact interval estimates, t-statistic based p-values and discussions regarding families of tests represent a result of the analytical choices made during secondary analysis.

Appendix B Extended analysis outputs

extended_analysis.py will generate six additional files including: extended_specifications.csv, classroom_influence.csv, wild_cluster_bootstrap.csv, gpa_interactions.csv, session_estimates.csv, and equivalence_sensitivity.csv. In addition to creating these files, extended_analysis.py will also create two additional graphical representations. All of these extensions represent exploratory analyses. Included in the Bootstrap Output File are both its Seed and Draw Count. Equivalence Margins are simply Illustrative and Retrospective. Therefore, pointwise confidence intervals should NOT be interpreted as providing simultaneous coverage.